

These are a few of my favorite things

Higher Segal spaces, Cobordism categories with topological Defects,
 W^* -categories and Bicommutant Categories

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Higher Structures: Recent developments and applications

$$\mathbf{Z}: \mathrm{Bord}_d^{\mathrm{decorations}} \rightarrow \mathcal{V}_d$$

The (∞, d) -category of QFTs with (topological) defect theories as Morphisms ¹

Joint with Severin Bunk

Fix a dimension d and a target category $\mathcal{V}$.

We work with smooth (higher) categories i.e. they are presheaves over $\mathbf{Cart}$.

- We define d -dimensional smooth oriented bordism k -category with l -levels of topological defects of increasing codimensions.

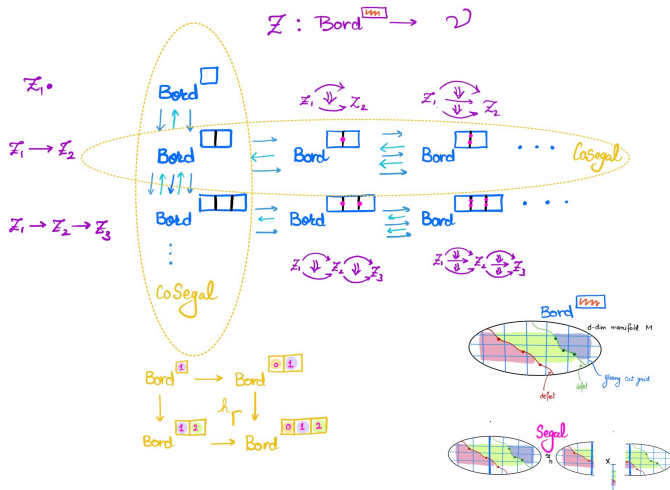
Here, $1 \leq k \leq d$ depending on how extended we want to be and $0 \leq l \leq d$ depending on the levels of symmetries or codimension of defects we are interested in.

- We show (modulo writing up) that this Bord is a k -uple Segal Space.
- Assuming $l = d$, we (hope to) show that our Bord with defects is l -uple coSegal making $\mathbf{Func}(\mathbf{Bord}, \mathcal{V})$ a d -category of QFTs, and theories with topological defects as morphisms.

¹Raindrops on roses and whiskers on kittens

The (∞, d) -category of QFTs with (topological) defects theories as Morphisms

Joint with Severin Bunk



Categorifying Von Neumann Algebras and their Modules

Joint with André Henriques and Dave Penneys

We want algebraic targets such that $\Omega^{d-1}(\mathcal{V}_d) = \mathbf{Vect}$ or $\mathbf{Hilb}$

Models for 2-Vect:

- $\mathbf{Mor}(\mathbb{E}_1(\mathbf{Vect}))$ that is $\mathbf{Mor}(\mathbf{Alg})$ 'equivalent to' $\mathbf{LinCat}$

Model for 3-Vect:

- $\mathbf{Mor}(\mathbb{E}_1(2\text{-Vect}))$ that is $\mathbf{Mor}(\mathbf{TensCat})$ 'equivalent to' $\mathbf{Lin\ 2-Cat}$.

Algebraic	Unitary Continuous
Vect	Hilb
$\mathbf{Mor}(\mathbf{Alg})$	$\mathbf{Mor}(\mathbf{vNAlg})$
LinCat	Complete W^* Cat
$\mathbf{Mor}(\mathbf{TensCat})$	$\mathbf{Mor}(\mathbf{BicommCat})$

Complete W^* - categories: 'categorified' Hilbert Spaces.

Bicommutant categories: 'categorified' of von Neumann algebras.

These models for 2-Hilb and 3-Hilb are inherently fully-dagger.

Categorifying Von Neumann Algebras and their Modules ²

Joint with André Henriques and Dave Penneys

For every von Neumann algebra A , there is a canonical Hilbert space ${}_A L^2 A_A$. The images of the left and right actions of A on $L^2 A$ are commutants of each other.

- **Fusion of Modules (Connes Fusion):**

$$M \boxtimes_A N = p_N \left(L^2(\text{End}_{A\text{-Mod}} (\overline{M} \oplus N)) \right) p_M$$

where p_N, p_M are the two projections

For every bicommutant category $\mathcal{T}$, there is a canonical W^* -category ${}_{\mathcal{T}} \mathcal{T}^{\text{abs}} {}_{\mathcal{T}}$. The images of the left and right actions of $\mathcal{T}$ on $\mathcal{T}^{\text{abs}}$ are commutants of each other.

- **Fusion of Modules ('categorified' Connes Fusion):**

$$\mathcal{M} \boxtimes_{\mathcal{T}} \mathcal{N} = p_{\mathcal{N}} \left(\text{End}_{\mathcal{T}\text{-Mod}} (\overline{\mathcal{M}} \oplus \mathcal{N}) \right)^{\text{abs}} p_M$$

²Bright copper kettles and warm woolen mittens

AQFT $\rightarrow$ FQFT for chiral Conformal Field Theories

my PhD thesis

- The AQFT incarnation of a unitary chiral CFT is a **conformal net on S^1** , which is a net of von Neumann algebras which is covariant under the action of $\text{Diff}^+(S^1)$.
- The FQFT incarnation of a chiral CFT is a **Segal chiral CFT** which is a functor $\text{Cob}_2^{\text{Conf}} \rightarrow \text{Mor}(\text{BicommCat})$.
- In my thesis, from a conformal net $\mathcal{A}$, I construct what the corresponding Segal chiral CFT assigns to 0 and 1 manifolds.
- This construction extends RT-type theories down to a point when the MTC assigned to the circle is realised as the representation category of a conformal net (or VOA).

AQFT $\rightarrow$ FQFT for chiral Conformal Field Theories ³

my PhD thesis

Point p	$\text{Sol}_p(\mathcal{A})$
Interval I	$\mathcal{A}(I)\text{-Mod}$ with the actions of $\text{Sol}_{\partial_- I}(\mathcal{A})$ and $\text{Sol}_{\partial_+ I}(\mathcal{A})$
Circle S	$\text{Rep}_S(\mathcal{A})^{\text{abs}}$

This assignment is **functorial** for gluing of arbitrary 1-manifolds.

We also show that $\text{Sol}(\mathcal{A})$ is a bicommutant category and $\mathcal{Z}_{\text{Drinfeld}}(\text{Sol}(\mathcal{A}))$ is $\text{Rep}(\mathcal{A})$.

³Brown paper packages tied up with strings

(tip of the iceberg of) Things I don't know yet ⁴

and would love to find out (if and) when I have my next job

- Extending our construction of orientated bordism categories with topological defects to bordisms with geometric structures.
- Interpretation of Gauging of Symmetries; Dualities; Symmetry TFT construction; homotopy groups of TQFTs in defect FQFT perspective
- The physical interpretation of $\text{Sol}(\mathcal{A})$? (as (non topological) defects)
- How much of the central charge of chiral CFT does $\text{Sol}(\mathcal{A})$ remember?
- Functorial constructions from chiral CFTs to full CFTs and back.
- Formulating universal properties and completions in the unitary world.
- Building and exploring the world of Bicommutant Categories.
See 75 problems on andreghenriques.com

⁴These are a few of my favourite things